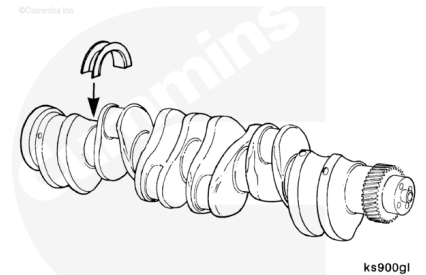


Trainings ▾

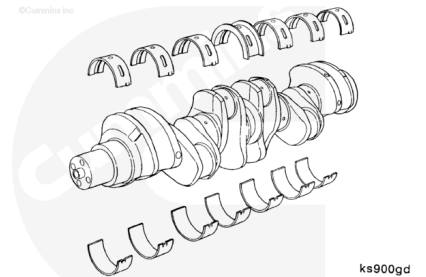
General Information

The crankshaft is a balanced, forged-steel unit. Four-cylinder engines have five main bearings. Six-cylinder engines have seven main bearings. The lower bearing shells are all the same. All of the upper bearing shells are also the same with the exception of the journal adjacent to the rear one. The next to last journal is fitted with a flanged upper bearing shell. The flanges control the end thrust of the crankshaft.



The crankshaft uses forged counterweights.

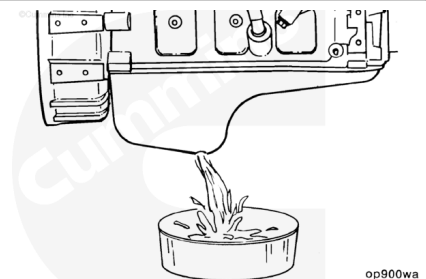
Oversize main bearings and thrust bearings are available for service. Cummins Inc., recommends regrinding **all** of the main or the connecting rod journals when one requires regrinding. Refer to the B Series Shop Manual, Bulletin 3666017.



Preparatory Steps

⚠ WARNING ⚠

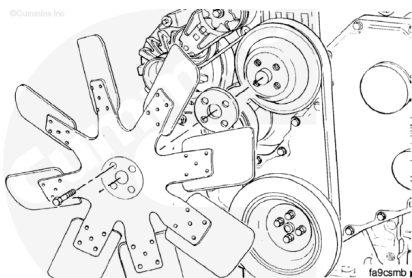
Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil.



Drain the lubricating oil; refer to Procedure 007-025 (</qs3/pubsys2/xml/en/procedures/40/40-007-025.html>).

Remove the fan pulley.

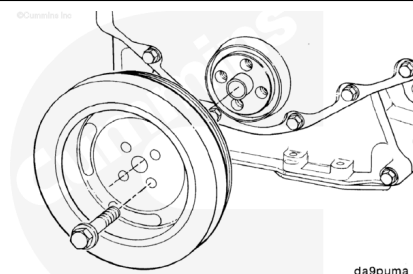
Refer to Procedure 008-039 (</qs3/pubsys2/xml/en/procedures/40/40-008-039-tr.html>).



Remove the vibration damper.

Refer to Procedure 001-052

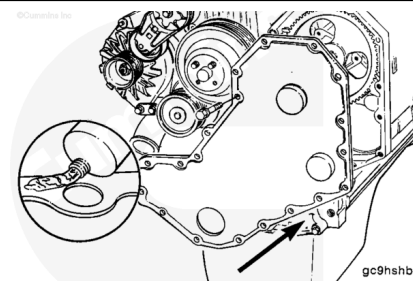
(/qs3/pubsys2/xml/en/procedures/40/40-001-052-tr.html).



Remove the front gear cover.

Refer to Procedure 001-031

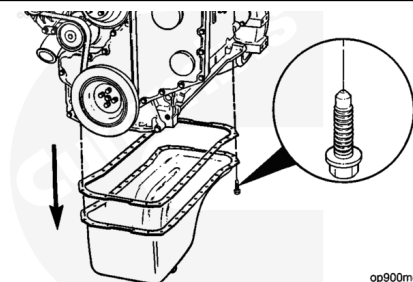
(/qs3/pubsys2/xml/en/procedures/40/40-001-031.html).



Remove the lubricating oil pan.

Refer to Procedure 007-025

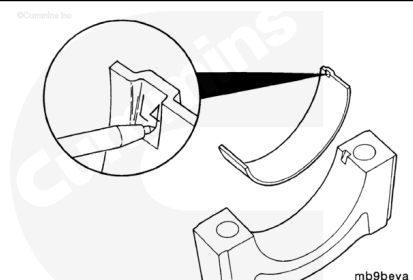
(/qs3/pubsys2/xml/en/procedures/40/40-007-025.html).



Remove the main bearings.

Refer to Procedure 001-006

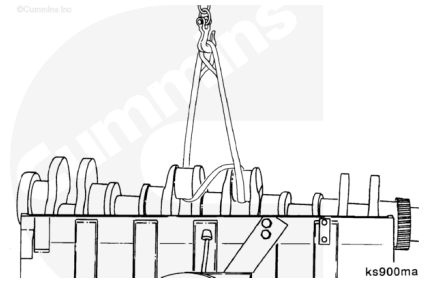
(/qs3/pubsys2/xml/en/procedures/40/40-001-006.html).



Remove

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.



Note : Lift the crankshaft straight up to reduce the possibility of damage to the crankshaft and cylinder block.

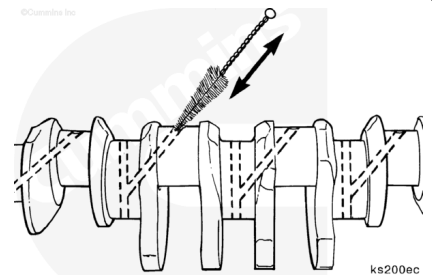
Install nylon lift sling, Part Number 3375957, around the Number 3 and Number 4 rod bearing journals.

Attach the sling to a hoist, remove the crankshaft.

Clean

Use fine crocus cloth to polish the machined surfaces.

Use a bristle brush to clean the oil drillings.



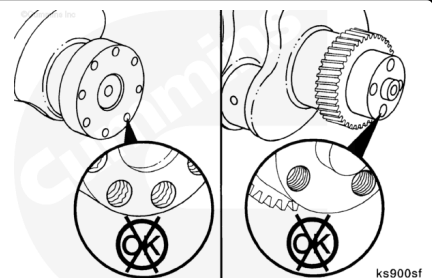
Inspect for Reuse

Inspect the threaded capscrew holes for damage.

Use one of the following methods to repair any threaded holes:

- Chase the threads
- Use the threaded insert kit, Part Number 3822709.

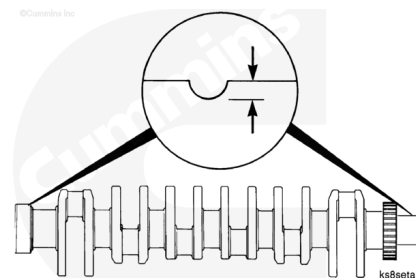
Note : If more than two threaded holes are damaged in one end, the crankshaft must be replaced.



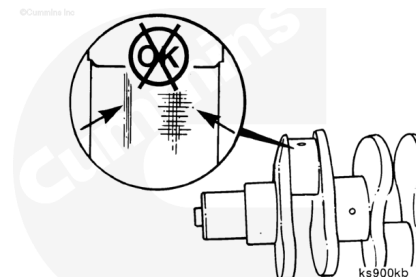
Measure the front and rear oil seal contact areas for a wear groove.

mm		in
0.25	MAX	0.010

Front and rear wear sleeves are available for a crankshaft that is worn beyond the limit.

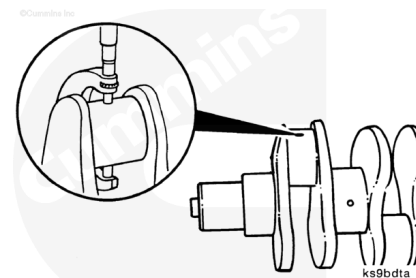


Inspect the crankshaft connecting rod and main journals for deep scoring, overheating, etc.



Use a micrometer to measure the connecting rod journal.

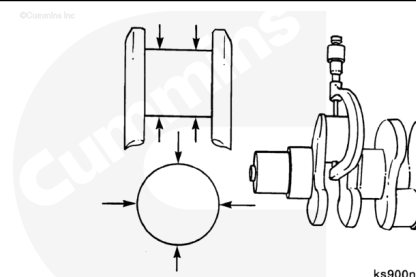
	mm		in
Standard	68.987	MIN	2.716
Machined 0.25 mm	68.737	MIN	2.706
Machined 0.50 mm	68.487	MIN	2.696
Machined 0.75 mm	68.237	MIN	2.687
Machined 1.00 mm	67.987	MIN	2.677



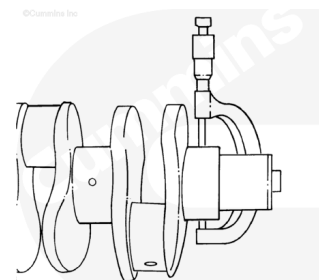
Bearing Clearance = Connecting rod inside diameter with bearing installed minus crankshaft journal diameter.

Maximum Bearing Clearance: 0.117 mm [0.0046 in].

	mm		in	
Out-of-Roundness		0.050	MAX	0.0020
Taper		0.013	MAX	0.0005



	mm		in
Standard	82.987	MIN	3.267
Machined 0.25 mm	82.737	MIN	3.257
Machined 0.50 mm	82.487	MIN	3.248

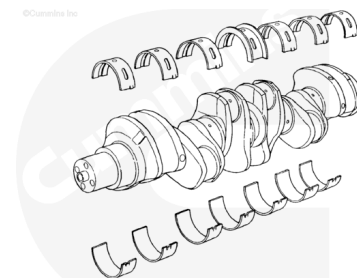


Bearing Clearance = Main bore diameter with bearing installed minus (-) crankshaft main journal diameter.

Maximum Bearing Clearance: 0.13 mm [0.005 in].

	mm		in	
Out-of-Roundness		0.05	MAX	0.0020
Taper		0.01	MAX	0.0004

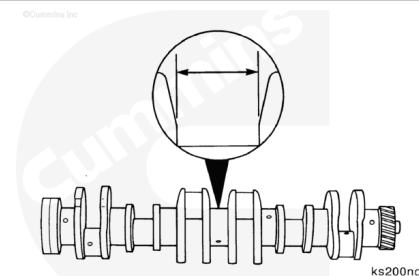
The following oversize service main and thrust bearings are available for crankshafts that have been machined undersize:



Thrust Size				
	Oversize	0.000	0.25 mm	0.50 mm
Journal	0.000	X	X	X
Diameter	0.25 mm	X	X	
	0.50 mm	X		X
	0.75 mm	X		
	1.00 mm	X		

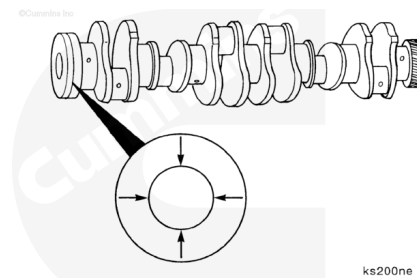
Measure the thrust face width.

mm		in
42.98	MIN	1.692
43.08	MAX	1.696



Measure the rear oil seal flange outside diameter.

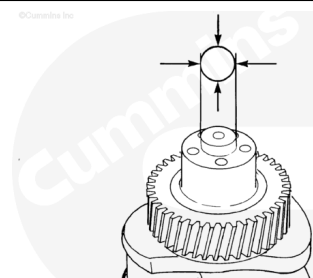
mm		in
129.98	MIN	5.117
130.03	MAX	5.119



ks200ne

Measure the damper pilot outside diameter.

mm		in
18.924	MIN	0.745
19.00	MAX	0.748

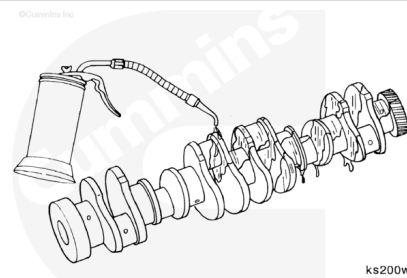


ks200nf

Install

Use a light preservative oil to lubricate the crankshaft to prevent rust.

Note : If the crankshaft is not going to be used immediately, protect the part with a plastic cover to prevent dirt from sticking to the oil.

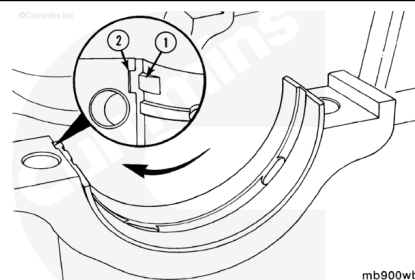


ks200wb

Install the upper main bearings.

Refer to Procedure 001-006

(/qs3/pubsys2/xml/en/procedures/40/40-001-006.html).

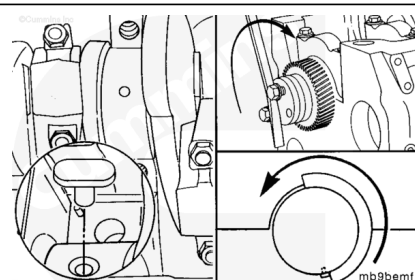


mb900wb

Install the upper thrust bearing.

Refer to Procedure 001-006

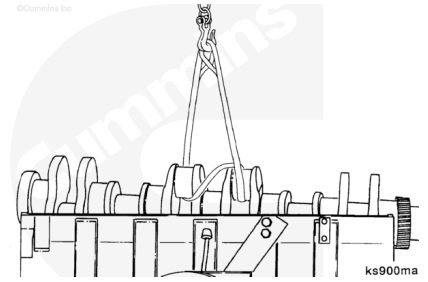
(/qs3/pubsys2/xml/en/procedures/40/40-001-006.html).



mb9bentf

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.



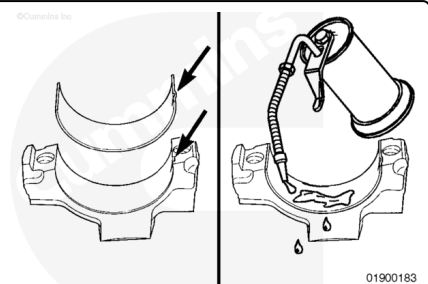
Use a hoist and nylon lift sling, Part Number 3375957.

Install the sling around the Number 3 and Number 4 connecting rod bearing journals.

Install the crankshaft.

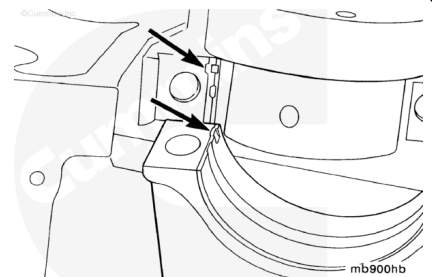
Install the lower main bearing.

Refer to Procedure 001-006
(</qs3/pubsys2/xml/en/procedures/40/40-001-006.html>).

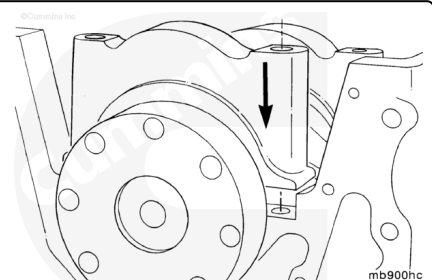


Note : The main bearing caps are numbered for location. Number 1 starts with the front of the cylinder block and the numbers **must** face the camshaft side of the engine.

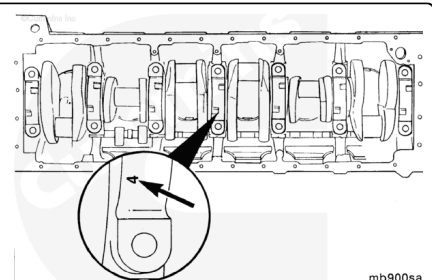
When installed correctly, the main bearing tangs should both be on the same side.



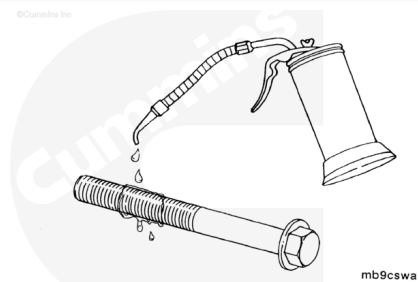
Position the main bearings and main bearing caps onto the crankshaft.



The thrust bearing journal does **not** have tangs or slots. Care **must** be taken to make sure the number stamped on the main bearing cap is positioned to the camshaft side of the engine.



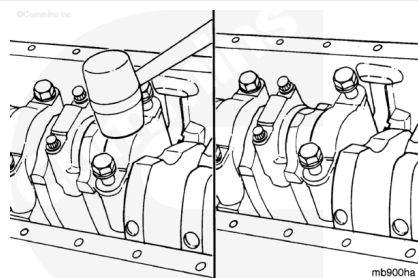
Lubricate the main bearing capscrew threads and underside of the capscrew head with clean lubricating engine oil.



Tap the main bearing cap gently into position.

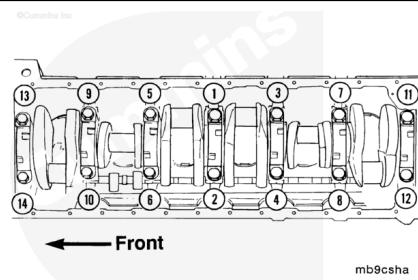
Note : Make sure the main bearing is still aligned with the cap.

When seated, the main bearing capscrews can be threaded in by hand.



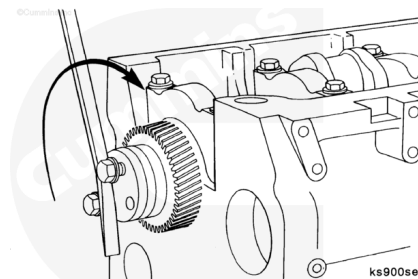
Tighten the main bearing capscrews evenly following the illustrated sequence.

Torque Value:	Step 1	60 n.m [44 ft-lb]
	Step 2	90 n.m [66 ft-lb]
	Step 3	Turn clockwise 90 degrees



Note : The crankshaft must rotate freely after the main bearings have been installed.

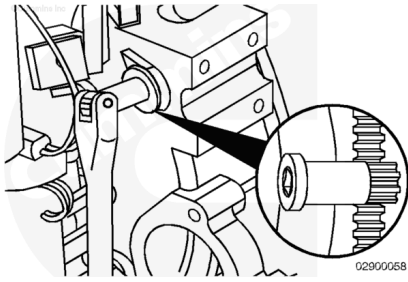
Inspect the main bearing installations and the size of the main bearings if the crankshaft will **not** rotate freely.



Rotation Check

Barring Tool, Part Number 3824591

The barring tool inserts into the flywheel housing and engages the flywheel ring gear. The crankshaft can then be rotated by hand using a 1/2-inch-drive ratchet or breaker bar.



Last Modified: 03-Mar-2004
